PTI STUDY GUIDE - MODULE 2: STRUCTURAL ASSESSMENT TOOLS

Introduction

Module 1 covered measurement techniques—the "what" and "how" of quantifying tree defects. Module 2 focuses on the **tools and technologies** used to gather that data. Professional tree assessment increasingly relies on specialized equipment to detect internal defects, quantify structural capacity, and predict failure risk.

This module covers four key assessment approaches:

1. **Sonic tomography** - mapping internal decay
2. **Resistograph** - drilling resistance profiles
3. **Pulling tests** - measuring structural response to loading
4. **Visual Tree Assessment (VTA)** - systematic observation protocols

Understanding when and how to use these tools—and their limitations—is essential for evidence-based tree management.

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1. SONIC TOMOGRAPHY

What Is Sonic Tomography?

Sonic tomography uses sound waves to create cross-sectional images of tree stems, revealing internal decay, cavities, and structural defects **non-invasively**.

Basic principle:

- Sound waves travel faster through sound wood than through decayed or hollow areas
- Sensors placed around the stem measure wave travel time
- Computer software processes data to create a visual "map" of internal condition

Common systems:

- **Arborsonic 3D** (Fakopp)
- **PiCUS Sonic Tomograph** (Argus Electronic)
- **Arbotom** (Rinntech)

How Sonic Tomography Works

Step 1: Sensor placement

- 8-12 sensors placed around stem circumference
- Spacing: roughly equal intervals around perimeter
- Height: typically 1-1.5m (breast height), or at defect location
- Sensors penetrate bark to contact sound wood

Step 2: Data collection

- Each sensor acts as both transmitter and receiver
- System sends sound pulse from each sensor
- All other sensors measure arrival time
- Creates matrix of travel times between all sensor pairs

Step 3: Data processing

- Software calculates expected vs actual travel times
- Slower travel = lower density = decay or cavity
- Creates color-coded tomogram (cross-sectional image)

Typical color scheme:

- **Dark green/brown**: Sound, dense wood
- **Yellow/orange**: Reduced density (early decay, crack, variation)
- **Red/white**: Very low density (advanced decay, cavity)

Interpreting Tomograms

Sound wood characteristics:

- Uniform color throughout
- Consistent density readings
- Smooth, circular cross-section

Decay indicators:

- Color change (usually brown → yellow → red)
- Irregular shapes
- Asymmetric patterns
- Percentage of cross-section affected

Cavity indicators:

- Bright red or white areas
- Sharp boundaries
- Can calculate remaining shell thickness (t/R ratio)

Important caveats:

False positives (appears decayed but isn't):

- Tension wood (natural in leans)
- Wet wood vs dry wood differences
- Cracks filled with water
- Species variation (some woods naturally less dense)

False negatives (decay missed):

- Very early-stage decay (wood structure intact)
- Certain fungal types (Kretzschmaria can be difficult to detect early)
- Small localized pockets between sensors
- Root decay (below measurement height)

When to Use Sonic Tomography

Good applications:

- Suspected internal decay without visible cavity
- Quantifying extent of known decay
- Mapping decay distribution (symmetric vs asymmetric)
- Calculating t/R ratio non-invasively
- Monitoring decay progression over time (repeat scans)
- Pre-construction surveys (valuable trees)

Poor applications:

- Routine screening (too expensive/time-consuming)
- Very small trees (sensor spacing issues)
- Very large trees (may need multiple scan heights)
- When visual cavity already provides sufficient information

- Root defects (different technique needed)

Practical Considerations

Cost: £££ (expensive equipment, requires trained operator)

Time: 20-45 minutes per tree (setup, scanning, interpretation)

Training: Requires understanding of wood structure, decay patterns, and software interpretation

Repeatability: Good—can rescan same location years later for comparison

Accuracy: High for detecting significant decay; less reliable for early-stage decay

Seasonal effects:

- Moisture content affects readings (prefer dry conditions)
- Frozen wood may give misleading results
- Best accuracy: late winter/early spring (consistent moisture)

Example Application

Scenario: Mature oak, 800mm diameter, suspected internal decay, Ganoderma fruiting body at base, high-value amenity location.

Tomography findings:

- Total radius: 400mm
- Decay cavity: 220mm radius (55% of cross-section)
- Sound shell: 180mm on all sides
- t/R ratio: 180 ÷ 400 = 0.45
- Decay distribution: Reasonably symmetric

Interpretation: Despite 55% cross-sectional decay, t/R of 0.45 indicates good residual structural capacity. Symmetric decay pattern more predictable than asymmetric. Recommend monitoring (repeat scan in 3-5 years) rather than immediate intervention.

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2. RESISTOGRAPH (Resistance Drilling)

What Is a Resistograph?

A **resistograph** is a drilling device that measures wood resistance as a fine needle penetrates the stem. The resulting resistance profile reveals internal structure, decay, and density variations.

Key features:

- Ultra-fine drill bit (1.5-3mm diameter)
- Constant drilling speed
- Measures drilling resistance continuously
- Produces graphical output (resistance vs depth)

Common systems:

- **IML Resistograph** (most common)
- **Sibert Resistograph**

How Resistograph Works

The drilling process:

1. Drill bit advances at constant speed (typically 40-60 cm/min)
2. Motor measures resistance (torque required to advance)
3. Higher resistance = denser/sounder wood
4. Lower resistance = decay, cavity, or natural low-density wood
5. Data recorded as graph: amplitude (resistance) vs distance

Reading the graph:

Healthy wood:

- Consistent amplitude throughout
- Regular pattern reflecting growth rings
- High baseline resistance

Decayed wood:

- Reduced amplitude
- Irregular pattern
- Sudden drops in resistance

Cavity:

- Resistance drops to near zero
- Flat line at baseline
- Sharp return to normal when exiting cavity

Growth ring pattern:

- Annual rings visible as slight oscillations
- Dense latewood = higher peaks
- Less dense earlywood = lower valleys
- Ring count confirms age

Interpreting Resistance Profiles

Pattern 1: Sound wood throughout

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|‾|‾|‾|‾|‾|‾|
```

Interpretation: Healthy stem, good structural capacity

Pattern 2: Decay pocket

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Interpretation: Localized decay, quantify depth and extent

Pattern 3: Hollow center

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Interpretation: Central cavity, measure shell thickness

Pattern 4: Progressive decay

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Interpretation: Decay advancing from one side, asymmetric loading risk

Pattern 5: Crack or split

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Interpretation: Brief drop = crack, returns to normal = crack hasn't opened fully

Quantifying Decay from Resistograph

Method 1: Visual estimation

- Identify regions of "normal" vs "reduced" resistance
- Mark boundaries on graph
- Estimate percentage of radius affected

Method 2: Threshold method

- Define threshold (e.g., 50% of maximum resistance)
- Measure length below threshold
- Express as % of total radius

Example:

- Total drill depth: 380mm (radius)
- Length below 50% threshold: 140mm
- Decay extent: 140 ÷ 380 = **37%**

Method 3: Area integration

- Calculate area under curve
- Compare decayed section to sound section
- More accurate but requires software

When to Use Resistograph

Good applications:

- Confirming suspected internal decay
- Measuring shell thickness in hollow trees
- Detecting decay in apparently sound stems
- Poles, posts, structural timbers (utility inspection)
- Tracking decay progression (repeat drilling at same location)
- Quick screening of multiple trees

Poor applications:

- Very small trees (drill creates proportionally large wound)
- When tomography would be more appropriate (complex decay patterns)
- As sole basis for high-consequence decisions (combine with other assessments)
- Friable, brittle decay (drill can't penetrate, gives false reading)

Advantages and Limitations

Advantages:

- Fast (5-15 minutes per tree)
- Portable, battery-powered
- Relatively inexpensive vs tomography
- Provides immediate results
- Minimal training required for basic interpretation
- Small wound (1.5-3mm hole)

Limitations:

- Single plane only (doesn't show 3D distribution)
- Invasive (creates wound, risk of introducing pathogens)
- Can't distinguish decay type
- Brittle wood breaks drill bit
- Misses decay between drill paths
- Requires multiple drillings for complete picture

Practical Considerations

Drill direction:

- Usually perpendicular to stem
- Consider lean direction (drill through thinnest shell)
- Multiple directions recommended for complete assessment (90° apart)

Drilling height:

- Breast height (1.3m) for consistency
- Or at visible defect location
- Or at loading point (e.g., below major branch)

Multiple drillings:

- Minimum 2 directions (perpendicular)
- 4 directions (N, E, S, W) for thorough assessment
- Avoid drilling directly into visible decay (spreads pathogens)

Wound management:

- No treatment needed (small wound compartmentalizes naturally)

- Avoid drilling trees with known active pathogens if possible
- Clean bit between trees (prevent pathogen transfer)

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3. PULLING TESTS (Static Load Tests)

What Is a Pulling Test?

A **pulling test** applies controlled lateral force to a tree, measuring its structural response. This simulates wind loading and reveals how close the tree is to failure.

What it measures:

- **Displacement:** How much the tree moves under load
- **Strain:** Deformation in wood fibers (compression and tension)
- **Load-displacement relationship:** Elastic vs plastic response
- **Failure threshold:** How much force would cause failure

Common systems:

- **TreeQinetic** (Argus Electronic)
- **Elastometer** (various manufacturers)
- **Manual pulling with inclinometers**

How Pulling Tests Work

Basic setup:

1. **Anchor point:** Secure anchor (vehicle, ground anchor, adjacent tree)
2. **Pulling rope:** Attached high in crown (typically 2/3 tree height)
3. **Force measurement:** Load cell in rope measures applied force
4. **Displacement sensors:**
    - Inclinometers on stem measure tilt
    - Root plate sensors detect lifting/rotation
5. **Data logging:** Computer records force and displacement continuously

Test procedure:

1. **Baseline measurement:** Record initial position (no load)

2. **Progressive loading:** Gradually increase pulling force
3. **Hold intervals:** Pause at intervals to observe response
4. **Maximum safe load:** Stop before causing damage (typically 50% of estimated failure load)
5. **Release:** Remove load, observe recovery

Target force:

- Typically aim for stem tilt of 0.25-0.5° (simulates wind load)
- Higher forces only if specific failure load needed
- Safety factor: never exceed 50% of calculated failure threshold

Interpreting Pulling Test Results

Key measurements:

1. Stem base deflection

- How much does base of stem tilt under load?
- Small deflection = stiff, strong structure
- Large deflection = flexible or weakened structure

2. Root plate movement

- Does root plate lift or rotate?
- No movement = secure anchorage
- Significant movement = root failure risk

3. Load-displacement curve shape

Healthy response (linear elastic):

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Force
 |    /
 |   /
 |  /
 | /
 |/__________
   Displacement
```

- Straight line
- Returns to original position when load removed

- Indicates elastic deformation only

Concerning response (plastic deformation):

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Force
 |     __--
 |   __/
 |  /
 | /
 |/_________
   Displacement
```

- Curve begins to flatten (yield point)
- Doesn't fully return to original position
- Indicates permanent deformation (failure initiating)

Critical response (near failure):

```
Force
 |    ___----
 |  _/
 | /
 |/_________
   Displacement
```

- Pronounced flattening
- Sudden acceleration of movement
- Test should stop immediately

Calculating Safety Factor

From pulling test data:

1. **Measure force at target tilt** (e.g., 0.25° = typical wind load)
    - Example: 5 kN to achieve 0.25° tilt
2. **Estimate failure load** (from curve shape or previous data)
    - Example: 18 kN (where curve shows yield behavior)
3. **Calculate safety factor:**
    - Safety Factor = Failure Load ÷ Operating Load
    - Example: 18 kN ÷ 5 kN = **3.6**

Interpretation:

- Safety Factor > 3.0: Structurally adequate
- Safety Factor 1.5-3.0: Monitor, consider intervention
- Safety Factor < 1.5: High failure risk, urgent management

When to Use Pulling Tests

Good applications:

- Trees with suspected root defects
- Recently developed lean (assess stability)
- After root damage (construction, severed roots)
- Validate tomography/resistograph findings
- High-value trees (justify retention or removal)
- Research (understanding failure mechanics)

Poor applications:

- Routine screening (too time/labor intensive)
- When obvious failure already present
- Very large trees (difficult to generate sufficient force)
- Protected species/locations (permission needed)
- When failure would be catastrophic (testing itself risks damage)

Advantages and Limitations

Advantages:

- **Functional test** - measures actual performance, not just anatomy
- **Whole tree response** - integrates all defects (root, stem, crown)
- **Quantitative** - produces numerical safety factors
- **Predictive** - identifies trees near failure threshold
- **Validates other methods** - confirms or refutes tomography/resistograph findings

Limitations:

- **Resource intensive** - requires equipment, time (2-4 hours), multiple operators
- **Expensive** - specialist equipment and expertise

- **Potentially damaging** - risk of causing failure during test
- **Access required** - need space for equipment, anchor points
- **Single loading direction** - may miss failure modes in other directions
- **Static test** - doesn't replicate dynamic wind loading fully

Safety Considerations

Critical safety requirements:

- Exclusion zone around tree (minimum 1.5× tree height)
- Test supervisor with authority to stop test
- Emergency plan if tree fails during test
- Load applied gradually with constant monitoring
- Stop immediately if unexpected behavior observed
- Never exceed 50% of estimated failure load

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4. VISUAL TREE ASSESSMENT (VTA)

What Is VTA?

Visual Tree Assessment (VTA) is a systematic methodology for identifying structural defects through careful observation. Developed by Prof. Claus Mattheck, VTA is based on the principle that **trees optimize their form for mechanical efficiency**, so deviations from ideal form indicate defects.

Core principle: "The tree doesn't lie" - external symptoms reliably indicate internal problems.

VTA Methodology

Step 1: Distant observation (10-20m)

Assess overall form and balance:

- **Crown asymmetry** - unbalanced crown suggests structural adaptation
- **Lean** - deviation from vertical indicates loading issues
- **Dead branches** - concentrated dieback suggests localized problems
- **Overall vigor** - stress indicators (foliage density, leaf size)

Step 2: Middle-distance observation (5-10m)

Examine structure in detail:

- **Branch architecture** - co-dominant stems, included bark, sharp angles
- **Branch distribution** - gaps in crown (recent failures?)
- **Wound wood** - evidence of previous damage
- **Fungal fruiting bodies** - location, type, number

Step 3: Close inspection (touching distance)

Detailed examination:

- **Bark condition** - cracks, seams, abnormal texture
- **Stem form** - ribs, flutes, swellings (reaction to stress)
- **Root collar** - girdling roots, decay, soil level changes
- **Cavities** - openings, extent, drainage
- **Cracks** - orientation, depth, active vs stable

Step 4: Below-ground (where accessible)

Root investigation:

- **Air spade** - expose root collar/major roots
- **Probing** - test for root decay
- **Root plate inspection** - lifting, cracks, mounding

VTA "Body Language" - Reading Tree Adaptations

Trees respond to stress by **adding material where needed**. VTA interprets these adaptations:

1. Ribs and flutes

- **What:** Vertical ridges along stem
- **Meaning:** Tree reinforcing compression zones
- **Implication:** Tree responding appropriately to loading (positive sign)
- **Concern:** If ribs suddenly stop → possible internal defect at that point

2. Swelling/bulging

- **What:** Localized enlargement
- **Meaning:** Tree adding material around weak point
- **Implication:** Attempted repair, but indicates underlying problem
- **Concern:** Disproportionate swelling = significant defect

3. Uniform taper

- **What:** Gradual diameter decrease from base to top
- **Meaning:** Optimal stress distribution
- **Implication:** Structurally sound (no localized weakness)
- **Concern:** Sudden change in taper = defect location

4. Lean with reaction growth

- **What:** Lean with eccentric growth rings (compression wood)
- **Meaning:** Chronic lean, tree adapted
- **Implication:** Stable if no progression
- **Concern:** Lean without reaction growth = recent, unstable

5. Crown asymmetry

- **What:** Unbalanced crown (more weight one side)
- **Meaning:** Tree compensating for root or stem defect opposite side
- **Implication:** May be stable, but investigate cause
- **Concern:** Progressive asymmetry = deteriorating condition

VTA Signs and Their Meanings

Fungal Fruiting Bodies:

Each fungal species indicates specific decay characteristics:

- **Ganoderma** - white rot, gradual progression, 5-15 year retention typical
- **Armillaria** - root/butt rot, variable progression, honey fungus
- **Kretzschmaria deusta** - brittle cinder, sudden failure risk, urgent concern
- **Meripilus giganteus** - root decay, fan-shaped brackets, reduces anchorage
- **Inonotus dryadeus** - butt rot specialist on oak, weeping brackets
- **Fomes fomentarius** - white rot, hoof-shaped, perennial brackets

VTA principle: Presence of fruiting body = significant decay already present (mycelium precedes fruiting by years)

Cracks and Seams:

- **Vertical cracks** - less concerning (parallel to grain, stress relief)

- **Horizontal cracks** - more concerning (perpendicular to grain)
- **Spiral cracks** - indicates twisting failure mode
- **Gaping cracks** - active failure in progress
- **Closed cracks with callus** - old damage, healing (less concern)

Cavities:

- **Well-compartmentalized** - rounded opening, callus growth (tree coping)
- **Expanding cavity** - uneven edges, no callus (progressing decay)
- **Drainage** - wet cavity interior (active decay likely)
- **Dry cavity** - may indicate arrested decay

Bark Condition:

- **Smooth, intact bark** - healthy, no trauma
- **Missing bark patches** - wound, potential decay entry
- **Raised bark** - swelling underneath (reaction growth or decay)
- **Longitudinal cracks** - normal in some species, concerning in others
- **Peeling bark** - dead tissue, dieback

The "Seven Sins" - Common Critical Defects

VTA identifies seven particularly concerning defect types:

1. **Included bark** - Bark trapped in union, prevents fusion
2. **Cavity at fork** - Decay in crotch, high failure risk
3. **Co-dominant stems** - Competing leaders, weak attachment
4. **Sharp branch angles** - <45° angle, poor attachment
5. **Root defects** - Girdling, decay, severed roots
6. **Brittle decay fungi** - Kretzschmaria, sudden failure
7. **Lean without adaptation** - Recent lean, no reaction wood

Combining VTA with Instrumental Assessment

VTA workflow:

1. **VTA identifies concern** (e.g., fungal fruiting body)
2. **Hypothesis formed** (e.g., internal decay suspected)

3. **Instrumental test confirms** (e.g., tomography quantifies extent)
4. **Integrated assessment** (VTA + instrument data → decision)

Example:

- VTA: Ganoderma bracket on oak, 15m height, stem bulging slightly
- Hypothesis: Internal decay, extent unknown, tree attempting compensation
- Resistograph: Confirms decay, measures shell thickness (t/R = 0.28)
- Decision: Adequate shell remaining, monitor annually, assess progression

VTA Limitations

What VTA does well:

- Identifies presence of defects
- Guides where to investigate further
- Interprets tree's adaptive responses
- Integrates multiple observations into diagnosis

What VTA cannot do:

- Quantify internal decay extent (needs instruments)
- Precisely predict failure timing (probability only)
- Detect very early-stage defects (no external symptoms yet)
- Replace professional judgment (requires experience)

Systematic VTA Inspection Protocol

Recommended sequence:

1. **Review records** (previous inspections, site history)
2. **Site context** (targets, constraints, environmental factors)
3. **Distant view** (overall form, major defects)
4. **Close approach** (structure, wounds, fungi)
5. **Detailed inspection** (stem, branches, root collar)
6. **Below-ground** (if warranted, if accessible)
7. **Document findings** (photos, measurements, notes)
8. **Hypothesis** (what defects are present?)

9. **Further investigation?** (instruments needed?)
10. **Risk assessment** (integrate all findings)

Time required:

- Basic VTA: 15-30 minutes per tree
- Detailed VTA: 45-90 minutes per tree
- Plus instrument time if needed

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5. SELECTING THE RIGHT TOOL

Decision Framework

Question 1: What are you trying to determine?

- **Presence of decay** → VTA first, then Resistograph to confirm
- **Extent of decay** → Tomography (full cross-section) or Resistograph (sampling)
- **Structural safety factor** → Pulling test
- **Root condition** → VTA + air spade, or pulling test for anchorage

Question 2: What's the consequence of being wrong?

- **High consequence** (targets, high value) → Multiple methods for confirmation
- **Moderate consequence** → Instrument + VTA usually sufficient
- **Low consequence** → VTA may be adequate

Question 3: What resources are available?

- **Budget £** → VTA only (skill-dependent)
- **Budget ££** → VTA + Resistograph
- **Budget £££** → VTA + Tomography or Pulling test

Question 4: Is repeat assessment needed?

- **Monitor progression** → Non-invasive preferred (Tomography, Pulling test)
- **One-time decision** → Resistograph acceptable

Tool Comparison Matrix

Tool	Cost	Time	Invasive	Best For	Limitations
VTA	£	30-60 min	No	Screening, defect identification	Doesn't quantify internal defects
Resistograph	££	15-30 min	Yes (small)	Confirming decay, quick assessment	Single plane, can miss distributed decay
Tomography	£££	30-90 min	No	Mapping decay distribution, t/R calculation	Expensive, complex interpretation
Pulling Test	£££	2-4 hours	No	Safety factor, whole-tree response	Resource intensive, safety concerns

Recommended Assessment Combinations

Scenario 1: Routine inspection

- Tool: VTA
- Rationale: Cost-effective screening, identifies trees needing further investigation

Scenario 2: Decay suspected (fruiting body present)

- Tools: VTA + Resistograph or Tomography
- Rationale: VTA identifies concern, instrument quantifies extent

Scenario 3: High-value tree, removal proposed

- Tools: VTA + Tomography + possible Pulling Test
- Rationale: Multiple lines of evidence justify decision, defensible if challenged

Scenario 4: Root damage from construction

- Tools: VTA + Air spade + Pulling Test
- Rationale: VTA screens, air spade reveals extent, pulling test measures impact on stability

Scenario 5: Progressive lean

- Tools: VTA + Pulling Test + repeat measurements
- Rationale: VTA distinguishes chronic vs recent, pulling test quantifies current safety factor, repeat assesses progression

SUMMARY: KEY PRINCIPLES

1. No Single Tool Is Perfect

- Each tool has strengths and limitations
- Combine methods for complete picture
- VTA should always be foundation

2. Instruments Provide Data, Not Decisions

- Tools measure conditions
- Professional interprets significance
- Context essential (species, loading, targets)

3. Match Tool to Question

- Define what you need to know
- Select tool(s) that answer that question
- Don't over-assess (wasted resources)
- Don't under-assess (increased risk)

4. Document Thoroughly

- Record what tools were used
- Include raw data (graphs, tomograms)
- Explain interpretation
- Forms evidence base for decisions

5. Know When to Call Specialists

- Complex decay patterns → arboricultural consultant
- High-consequence decisions → second opinion
- Novel situations → research literature or specialist advice
- Legal implications → ensure defensible assessment

PRACTICE EXERCISES

Exercise 1: Tool Selection

Scenario: Oak tree, 18m height, 800mm diameter, small Ganoderma bracket at base (100mm diameter), tree overhangs children's play area.

Question: What assessment tool(s) would you recommend? Justify your choice.

Answer: VTA + Sonic Tomography recommended.

- VTA confirms decay indicators, checks for other defects
- Tomography needed due to high consequence (children), quantifies decay extent for defensible decision
- Resistograph insufficient (single plane, high-consequence decision needs fuller picture)

Exercise 2: Resistograph Interpretation

Resistograph profile shows:

- First 150mm: High resistance (70-85% amplitude)
- 150-180mm: Gradual decline to 30% amplitude
- 180-280mm: Low resistance (20-30% amplitude)
- 280-320mm: Gradual increase to 60% amplitude
- 320-380mm: High resistance (75-90% amplitude)

Tree radius: 380mm

Question: Calculate approximate t/R ratio. What does this indicate?

Answer:

- Sound shell one side: 150mm
- Sound shell other side: 380 - 320 = 60mm
- Average shell: (150 + 60) ÷ 2 = 105mm
- t/R = 105 ÷ 380 = **0.28**
- **Interpretation:** Moderate concern level, asymmetric decay (one side much thinner), requires detailed assessment and management consideration.

Exercise 3: VTA Observation

Tree shows: lean of 8°, reaction wood visible on increment core (wide rings on tension side), no soil disturbance, well-developed buttresses.

Question: Is this a concern? Explain your reasoning.

Answer: Lower concern - chronic adapted lean.

- Reaction wood indicates long-term adaptation
- No soil disturbance = roots not failing
- Buttresses show structural compensation
- Recommendation: Monitor lean progression (is 8° increasing?), but tree is coping with current condition.

NEXT MODULE

Module 3: Cable & Support Systems
Covers installation standards, load-sharing mechanics, and when cabling is effective vs when it's inappropriate.

This study guide provides foundational knowledge for PTI training. These assessment tools generate the data you'll apply in practice scenarios.